Emanuel Tewolde

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SELECTED EDUCATION

Carnegie Mellon University Aug. 2022 – ca. Dec. 2027
PhD in Computer Science Pittsburgh, USA

- Foundations of Cooperative AI Lab (FOCAL), Advisor: Vincent Conitzer
- Focus: Algorithmic Game Theory for Agentic AI, LLM Agents, Reinforcement Learning

Carnegie Mellon University Aug. 2022 – May 2025
MSc in Computer Science — (research-focused degree) Pittsburgh, USA

Imperial College London Oct. 2020 – Oct. 2021
MSc in Pure Mathematics London, United Kingdom

- GPA (UK): 83.8%, Distinction → Ranked 1st out of 28 Graduates

Technical University (TU) of Darmstadt Oct. 2016 – Oct. 2019
BSc in Mathematics with a minor in Economics and Business Administration Darmstadt, Germany

- GPA (Germany): 1,27

INDUSTRY AND RESEARCH EXPERIENCES

Carnegie Mellon University Aug. 2022 – ca. Dec. 2027
Graduate Research Assistant Pittsburgh, USA

Fujitsu Jul. 2026 – Dec. 2026
AI Security Research Intern within Fujitsu Research of America Pittsburgh, USA

Meta (contracting via Magnit) Sep. 2025 – Dec. 2025
Research Engineer at FAIR *within Meta Superintelligence Labs* London, United Kingdom

- Contributed to LLM Coding Agents for AI Research (AIRA) by designing ML research tasks, spearheading the integration of RL and Game Theory environments, and conducting large-scale evaluations of agent capabilities.

University of Oxford Summers of 2022–2025
Extended Research Visits Oxford, United Kingdom

Fraunhofer Institute for Energy Economics and Energy System Technology Apr. 2020 – Jul. 2020
Research Internship, Deep Learning on Graphs representing Energy Distribution Networks Kassel, Germany

Fraunhofer Institute for Energy Economics and Energy System Technology May 2019 – Oct. 2019 resp.
Research Assistant, Machine Learning on Time Series for Wind Turbine Maintenance Kassel, Germany

SELECTED SCHOLARSHIPS, GRANTS, AND AWARDS

- WINE Best Student Paper Award (Top 2 out of 503 submissions) 2026
- 2x Best Paper Honorable Mentions at AI4GOOD@ICML26 (Top 6 out of 539 submissions) 2026
- AAAI Best Poster Award (Top 3 out of 12,957 submissions) 2025
- PhD Fellowship, Cooperative AI Foundation 2025 – present
- Research Grant, Cooperative AI Foundation, $6750 2022
- Scholarship, German Scholarship Foundation (Studienstift. des Deutschen Volkes), ∼22,000€ 2020 – 2022
- Scholarship, Hessen-Wisconsin Exchange Program, ∼$10,000 2018

PUBLICATIONS

$(\alpha\beta)$: alphabetical ordering, $^{=}$: equal contribution, $^{+}$: equal advising

Journal Papers

- Alisia Lupidi$^{=}$, Bhavul Gauri$^{=}$, Thomas Simon Foster$^{=}$, Bassel Al Omari$^{=}$, Despoina Magka$^{=}$, ... (11 more authors) ..., **Emanuel Tewolde**, ... (18 more authors) ..., Jakob Foerster, Yoram Bachrach. AIRS-Bench: a Suite of Tasks for Frontier AI Research Science Agents. In *Transactions on Machine Learning Research (TMLR)*, 2026.

Refereed Conference Papers

- Ratip Emin Berker$^{=}$, Ioannis Anagnostides$^{=}$, **Emanuel Tewolde**, Brian Hu Zhang, Tuomas Sandholm, Vincent Conitzer. What You Earn from What You Learn: Characterization and Complexity of Value of Information. To appear in *Web and Internet Economics (WINE)*, 2026. **Best Student Paper Award** (Top 2 out of 503 submissions).
- Steffen Backmann, David Guzman Piedrahita, Terry Jingchen Zhang, **Emanuel Tewolde**, Rada Mihalcea, Bernhard Schölkopf, Zhijing Jin. When Ethics and Payoffs Diverge: LLM Agents in Morally Charged Social Dilemmas. To appear in *Asia-Pacific Chapter of the Association for Computational Linguistics (AACL) and International Joint Conference on Natural Language Processing (IJCNLP)*, 2026.
- $(\alpha\beta)$ Ioannis Anagnostides, Maria-Florina Balcan, Kiriaki Fragkia, Tuomas Sandholm, **Emanuel Tewolde**, Brian Hu Zhang. The Complexity of Equilibrium Refinements in Potential Games. In *ACM Conference on Economics and Computation (EC)*, 2026.
- **Emanuel Tewolde**$^{=}$, Xiao Zhang$^{=}$, David Guzman Piedrahita, Vincent Conitzer^{+}, Zhijing Jin^{+}. CoopEval: Benchmarking Cooperation-Sustaining Mechanisms and LLM Agents in Social Dilemmas. In *International Conference on Machine Learning (ICML)*, 2026. **Best Paper Honorable Mention** at AI4GOOD@ICML26 (Top 6 out of 539 submissions).
- Ioannis Anagnostides, **Emanuel Tewolde**, Brian Hu Zhang, Ioannis Panageas, Tuomas Sandholm, Vincent Conitzer. Convergence of Regret Matching in Potential Games and Constrained Optimization. In *International Conference on Learning Representations (ICLR)*, 2026.
- Ratip Emin Berker$^{=}$, **Emanuel Tewolde**$^{=}$, Vincent Conitzer, Mingyu Guo, Marijn Heule, Lirong Xia. On the Edge of Core (Non-)Emptiness: An Automated Reasoning Approach to Approval-Based Multi-Winner Voting. In *Association for the Advancement of Artificial Intelligence (AAAI)*, **Oral**, 2026.
- Chandler Smith, Marwa Abdulhai, ... (15 more authors) ..., **Emanuel Tewolde**, ... (66 more authors) ..., Jose Hernandez-Orallo, Joel Z Leibo. Evaluating Generalization Capabilities of LLM-Based Agents in Mixed-Motive Scenarios Using Concordia. In *Neural Information Processing Systems (NeurIPS) – Datasets and Benchmarks Track*, 2025.
- Brian Hu Zhang$^{=}$, Ioannis Anagnostides$^{=}$, **Emanuel Tewolde**, Ratip Emin Berker, Gabriele Farina, Vincent Conitzer, and Tuomas Sandholm. Learning and Computation of Φ-Equilibria at the Frontier of Tractability. In *ACM Conference on Economics and Computation (EC)*, 2025.
- Brian Hu Zhang$^{=}$, Ioannis Anagnostides$^{=}$, **Emanuel Tewolde**, Ratip Emin Berker, Gabriele Farina, Vincent Conitzer, and Tuomas Sandholm. Expected Variational Inequalities. In *International Conference on Machine Learning (ICML)*, **Oral** (Top 1.0% of submissions), 2025.
- **Emanuel Tewolde**, Brian Hu Zhang, Caspar Oesterheld, Tuomas Sandholm, and Vincent Conitzer. Computing Game Symmetries and Equilibria That Respect Them. In *Association for the Advancement of Artificial Intelligence (AAAI)*, **Oral** (Top 4.6% of submissions), 2025. **Best Poster Award** (Top 3 out of 12,957 submissions).
- Ratip Emin Berker, **Emanuel Tewolde**, Ioannis Anagnostides, Tuomas Sandholm, Vincent Conitzer. The Value of Recall in Extensive-Form Games. In *Association for the Advancement of Artificial Intelligence (AAAI)*, **Oral** (Top 4.6% of submissions), 2025.
- **Emanuel Tewolde**, Brian Hu Zhang, Caspar Oesterheld, Manolis Zampetakis, Tuomas Sandholm, Paul W. Goldberg, and Vincent Conitzer. Imperfect-Recall Games: Equilibrium Concepts and Their Complexity. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
- **Emanuel Tewolde** and Vincent Conitzer. Game Transformations That Preserve Nash Equilibria or Best-Response Sets. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
- $(\alpha\beta)$ Vincent Conitzer, Rachel Freedman, Jobst Heitzig, Wesley H. Holliday, Bob M. Jacobs, Nathan Lambert, Milan Mossé, Eric Pacuit, Stuart Russell, Hailey Schoelkopf, **Emanuel Tewolde**, and William S. Zwicker. Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback. In *International Conference on Machine Learning (ICML) – Position Paper Track*, 2024.

- **Emanuel Tewolde**, Caspar Oesterheld, Vincent Conitzer, and Paul W. Goldberg. The Computational Complexity of Single-Player Imperfect-Recall Games. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2023.

Working Papers

- Ida Mattsson, Ratip Emin Berker, **Emanuel Tewolde**, Ioannis Anagnostides, Kiriaki Fragkia, Xinyu Liu, Alexander Heckett, Jiayuan Liu, Eden Hartman, Vojtěch Kovařík, Sven Neth, Maria-Florina Balcan, Tuomas Sandholm, Vincent Conitzer. Fundamental Impossibility Results in AI Safety: Trading Off between Desiderata. *Preprint*.
- Jiayuan Liu, Shiyi Du, Tianqin Li, **Emanuel Tewolde**, Keyang Xuan, Vincent Conitzer. Coopshift: Post-Training Amplifies The Memory Curse In LLM Cooperation. *Preprint*.
- Akash Kundu[=], **Emanuel Tewolde**[=], Ratip Emin Berker, Samuel F. Brown, Vincent Conitzer. Do LLMs Take Care of Their Own? Similarity Signals Can Induce Cooperation. *Preprint*. **Best Paper Honorable Mention** at AI4GOOD@ICML26 (Top 6 out of 539 submissions).
- Claire Jin, Andy Liu, **Emanuel Tewolde**, Vincent Conitzer. Evaluating Strategic Reasoning in LLMs in Simulation-Augmented Trust Games. *Preprint*.
- Jiayuan Liu[=], Tianqin Li[=], Shiyi Du[=], Xin Luo[=], Haoxuan Zeng, **Emanuel Tewolde**, Tai Sing Lee, Tonghan Wang, Carl Kingsford, Vincent Conitzer. The Memory Curse: How Expanded Recall Erodes Cooperative Intent in LLM Agents. *Preprint*.
- **Emanuel Tewolde**, Brian Hu Zhang, Ioannis Anagnostides, Tuomas Sandholm, Vincent Conitzer. Decision Making under Imperfect Recall: Algorithms and Benchmarks. *Preprint*.

Manuscripts

- Alexis Audran-Reiss, Jordi Armengol-Estapé, ... (12 more authors) ..., **Emanuel Tewolde**, ... (3 more authors) ..., Tatiana Shavrina, Yoram Bachrach. What Does It Take to Be a Good AI Research Agent? Studying the Role of Ideation Diversity. 2025.

RELEVANT SKILLS

Software: Python ★★★★★, High-Performance & Distributed Computing ★★★★⋆, PyTorch ★★★★⋆, TensorFlow ★★★⋆⋆, Gym ★★★★⋆
Languages: German ★★★★★, English ★★★★★, Amharic ★★★⋆⋆

Research Mentorship

- Xiao Zhang 2025 – present
  Project: Cooperation Mechanisms for LLM Agents — *Best Paper Honorable Mention, AI4GOOD@ICML26*
  Started as: Undergrad at UToronto; now: SWE at Google
- Akash Kundu 2026
  Project: Similarity-based Cooperation in LLM Agents — *Best Paper Honorable Mention, AI4GOOD@ICML26*
  Started as: Cooperative AI Research Fellow and Undergrad at Heritage Institute of Technology; now: Evaluations Engineer at Apart Research

TEACHING EXPERIENCE

- Reading Group Lead in Preparation for the Cooperative AI Summer School 2025
- Graduate Teaching Assistant, Carnegie Mellon University 2024 – 2025
  Sole TA for 15-784 Cooperative AI and 17-759 Advanced Topics in Machine Learning and Game Theory
- Instructor for new Teaching Assistants, TU of Darmstadt 2019 – 2020
  Taught and supervised 5 – 10 new TAs per semester for the mathematics department
- Undergraduate Teaching Assistant, TU of Darmstadt 2017 – 2019
  Introduction to Algebra, Analysis 2, Analysis 1

RESEARCH COMMUNICATION

Blog Posts

- A Science of Cooperation Infrastructure: Enabling Rational Cooperation Among LLM Agents. *Cooperative AI Foundation Blog* — Sept. 2026
- Leveraging Symmetries in Strategic Games. *CMU CSD PhD Blog* — May 2025

Invited Talks

- EconCS Seminar, Harvard University — Oct. 2026
- Algorithms & Complexity Seminar, Massachusetts Institute of Technology — Oct. 2026
- ML/AI Seminar, Carnegie Mellon University — Oct. 2026
- Guest Lecture for 15-784 Foundations of Cooperative AI, Carnegie Mellon University — Oct. 2026
- Thomas Kesselheim's and Heiko Röglin's Lab, University of Bonn — July 2026
- Georgia Koppe's Lab Heidelberg University — July 2026
- Max Planck Institute for Informatics (Saarbrücken) — July 2026
- AI Center Seminar, École Polytechnique Fédérale de Lausanne — July 2026
- Decision Sciences and Systems, Technical University of Munich — July 2026
- Google DeepMind London — June 2026
- Kaiqing Zhang's Lab, University of Maryland, College Park — Mar. 2026
- EconCS Seminar, Carnegie Mellon University — Feb. 2026
- CAIRF Retreat, AI Safety South Africa — Feb. 2026
- FAIR Meta London — Dec. 2025
- Distributed AI Seminar, Kings College London — Dec. 2025
- Autonomous Reasoning & Learning Group, Imperial College London — Jul. 2025
- Optimization Seminar, Technical University of Darmstadt — Jun. 2025
- EconCS Seminar, Carnegie Mellon University — Feb. 2025
- FAR.AI — Nov. 2024
- Guest Lecture for 17-759 Adv. Top. in Machine Learning and Game Theory, Carnegie Mellon University — Nov. 2024
- Foundation and Language Model (FLAME) Center, Carnegie Mellon University — Sep. 2024
- Algorithmic Game Theory Seminar, University of Oxford — Jun. 2024
- Google DeepMind London — May 2024
- Fei Fang's AI and Social Good Lab, Carnegie Mellon University — Apr. 2024

FURTHER HIGHER EDUCATION

- Multiple Courses of the MSc in International Economics & Economic Policy at Goethe University Frankfurt — 2019 – 2020
- MSc Specialization in Mathematical Optimization at the Technical University of Darmstadt — 2019 – 2020
- Exchange Semester at the University of Wisconsin-Milwaukee — 2018
- Taking a Mathematics Course at Goethe University Frankfurt as a High School Student — 2015

SELECTED VOLUNTEERING

- Mentoring (non-research) of more junior PhD students at Carnegie Mellon University — 2023 – present
- Co-organizing the visiting days and introductory course for admitted and first-year PhD students resp. — 2023 – 2025
- Academic Representative for Pure Mathematics at Imperial College London — 2020 – 2021
- Coordinating Outreach Events for the International Office at the TU of Darmstadt — 2019 – 2020
- Hessen-Student-Ambassador in Milwaukee — 2018